Intelligent Systems Coursework Report

Student ID:

Name:

Project Title: Customer Service Agent Intelligent System

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1. Application and Workflow Selection

1.1 Introduction and Motivation

The fast development of online services has multiplied the number of interactions with customers in different areas. Both businesses and customers desire quick and effective service and cost reduction in service provision and quality service enhancement. The human-based customer service that has been traditionally used is usually limited by features like slow response, inconsistency when dealing with the requests, and ineffectiveness of prioritizing the essential problems (Thorat, 2025). The force behind designing the Customer Service Agent intelligent system is to automate and streamline dealing with customer inquiries. Using the power of AI agents that can analyze queries, categorize problems, gauge customer mood, and suggest the right answers, companies can be guaranteed timely resolutions and enhanced customer satisfaction levels. Additionally, such a system can reduce the reliance on the human agents for routine tasks, and they will have enough time to concentrate on the complex situations when critical thinking and personal interaction are needed (Sapkota, Roumeliotis and Karkee, 2025).

1.2 Explanation of the Workflow with Diagram

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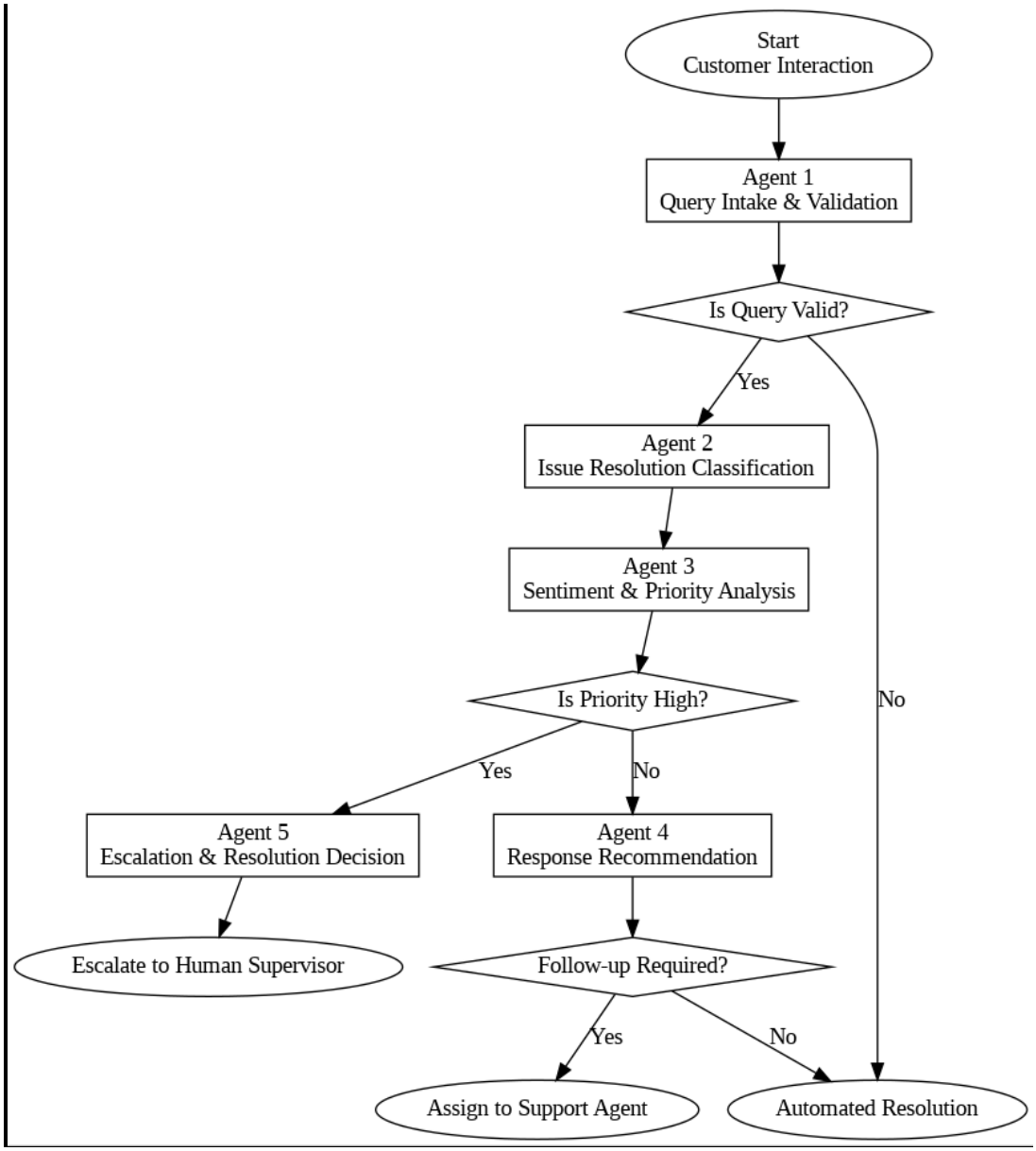
Start
Customer Interaction
Agent 1
Query Intake & Validation
Is Query Valid?
Yes
No
Agent 2
Issue Resolution Classification
Agent 3
Sentiment & Priority Analysis
Is Priority High?
Yes
No
Agent 5
Escalation & Resolution Decision
Agent 4
Response Recommendation
Escalate to Human Supervisor
Follow-up Required?
Yes
No
Assign to Support Agent
Automated Resolution

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C: > Users > nitee > Downloads > Intelligent_system_Bella.ipynb > # Install Kaggle

Generate + Code + Markdown Run All ···

```python
# Install Kaggle
!pip install -q kaggle

KAGGLE_USERNAME = "your_kaggle_username"
KAGGLE_KEY = "your_kaggle_api_key"

import os

# Set environment variables
os.environ["KAGGLE_USERNAME"] = KAGGLE_USERNAME
os.environ["KAGGLE_KEY"] = KAGGLE_KEY

# Create directory for dataset
os.makedirs("customer_service_interaction_dataset", exist_ok=True)

# Download the dataset
!kaggle datasets download -d zadafiyabhrami/customer-service-ineraction-dataset -p customer_service_inte

# Unzip the dataset
!unzip -o customer_service_interaction_dataset/*.zip -d customer_service_interaction_dataset

# List downloaded files
os.listdir("customer_service_interaction_dataset")
```

[1]

```
Dataset URL: https://www.kaggle.com/datasets/zadafiyabhrami/customer-service-ineraction-dataset
License(s): Community Data License Agreement - Sharing - Version 1.0
Downloading customer-service-ineraction-dataset.zip to customer_service_interaction_dataset
  0% 0.00/601k [00:00<?, ?B/s]
100% 601k/601k [00:00<00:00, 998MB/s]
Archive:  customer_service_interaction_dataset/customer-service-ineraction-dataset.zip
  inflating: customer_service_interaction_dataset/customer_service_interactions.csv

['customer-service-ineraction-dataset.zip',
 'customer_service_interactions.csv']
```

```python
import pandas as pd
import os
```

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
```

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```python
# Load dataset
data_path = "customer_service_interaction_dataset/customer_service_interactions.csv"
df = pd.read_csv(data_path)

# Input features
features = [
    "handled_by",
    "region",
    "device_type",
    "customer_sentiment",
    "query_type",
    "follow_up_required",
    "response_time_min",
    "interaction_cost_usd"
]

# Target column with stronger prediction power
target_column = "resolution_status"

# Keep required columns and remove missing values
df = df[features + [target_column]].dropna()

# Encode categorical variables
label_encoders = {}
for column in df.columns:
    if df[column].dtype == "object":
        encoder = LabelEncoder()
        df[column] = encoder.fit_transform(df[column])
        label_encoders[column] = encoder

# Split features and target
X = df[features]
y = df[target_column]
```

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```
Files in dataset directory:
['customer-service-ineraction-dataset.zip', 'customer_service_interactions.csv']
Dataset loaded successfully
Dataset shape:
(10000, 13)
Column names:
['interaction_id', 'timestamp', 'query_type', 'handled_by', 'response_time_min', 'resoluti
Sample records:
                      interaction_id   timestamp        query_type  \
0  d3053ddd-7ea3-447d-9c52-2d2e14ff40fe  2024-04-12  Billing Question
1  4a2c75d4-8687-4350-a595-f15e82720fbd  2024-04-26    Password Reset
2  e9f26019-4e13-4470-8068-81a4dfa5cf83  2024-02-18  Shipping Inquiry
3  b6e712b5-aa16-4ff0-a9cd-90f3cb11fe84  2024-04-17    Account Update
4  5c6ca2ba-8c3d-4567-a336-5c0bb6e0978d  2024-03-24  Billing Question

    handled_by  response_time_min resolution_status  customer_sentiment  \
0   AI Chatbot               1.88          Resolved                0.12
1   AI Chatbot               1.09          Resolved                0.14
2   AI Chatbot               1.13           Pending                0.26
3   AI Chatbot               1.12          Resolved                0.33
4  Human Agent               9.85         Escalated                0.87

   customer_satisfaction  interaction_cost_usd  \
0                      3                  2.81
1                      4                  0.13
2                      2                  2.75
3                      3                  4.42
4                      2                  5.26
```

```
                         customer_id       region device_type  \
0  06f211ac-4d8e-4a98-988c-be2d96c45b2f     Delaware      Mobile
1  bad3cd0f-24e7-4999-8e76-ccd77483db4e       Hawaii      Tablet
2  fd882d13-6f84-4969-b36a-9958b58b2380     Maryland      Mobile
3  fba76a29-dd57-4cfa-98a5-ce9d7f3c82f7  Connecticut      Mobile
4  daee835d-3f72-493e-beff-b2026da9ab53  Mississippi      Mobile

   follow_up_required
0               False
1               False
2               False
3                True
4                True
Missing values in each column:
interaction_id           0
timestamp                0
query_type               0
handled_by               0
response_time_min        0
resolution_status        0
customer_sentiment       0
customer_satisfaction    0
interaction_cost_usd     0
customer_id              0
region                   0
device_type              0
follow_up_required       0
dtype: int64
```

```
Basic statistics of numeric columns:
       response_time_min  customer_sentiment  customer_satisfaction  \
count       10000.000000        10000.000000            10000.00000
mean            4.302933            0.350261                3.28630
std             3.482249            0.296517                1.00858
min             0.200000           -0.750000                1.00000
25%             1.450000            0.150000                3.00000
50%             2.550000            0.350000                3.00000
75%             7.072500            0.550000                4.00000
max            17.150000            1.000000                5.00000

       interaction_cost_usd
count          10000.000000
mean               5.022155
std                2.504918
min               -2.640000
25%                3.010000
50%                5.040000
75%                7.040000
max               12.890000
```

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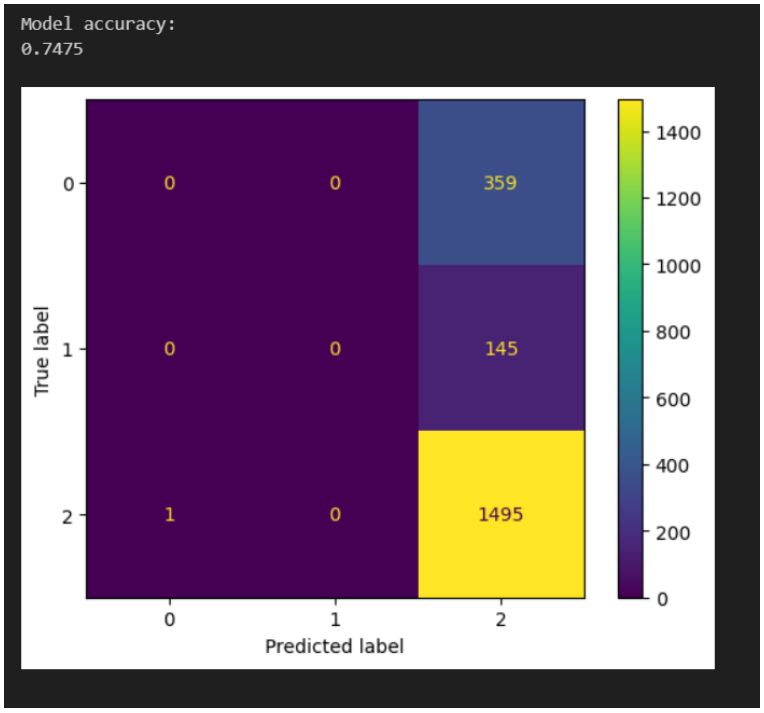

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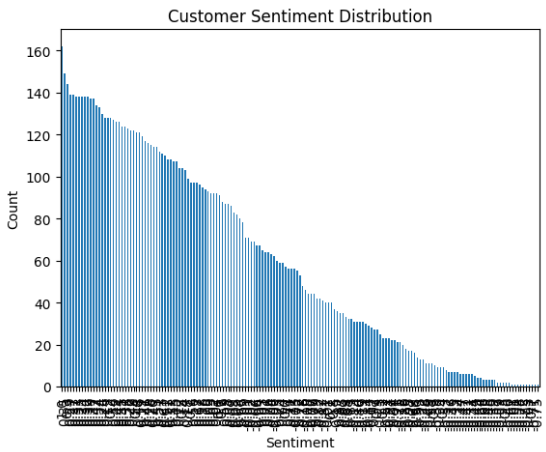

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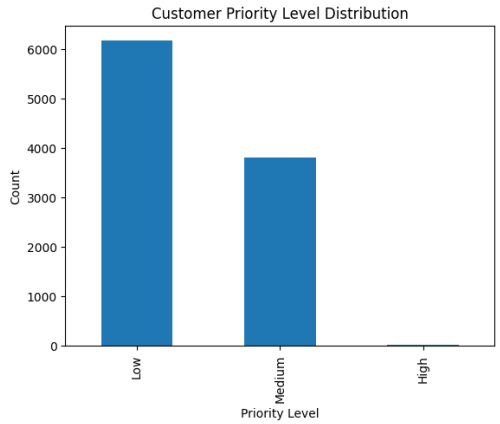


```
Sample sentiment and priority data:
   customer_sentiment  response_time_min priority_level
0                0.12               1.88           Low
1                0.14               1.09           Low
2                0.26               1.13           Low
3                0.33               1.12           Low
4                0.87               9.85        Medium
```

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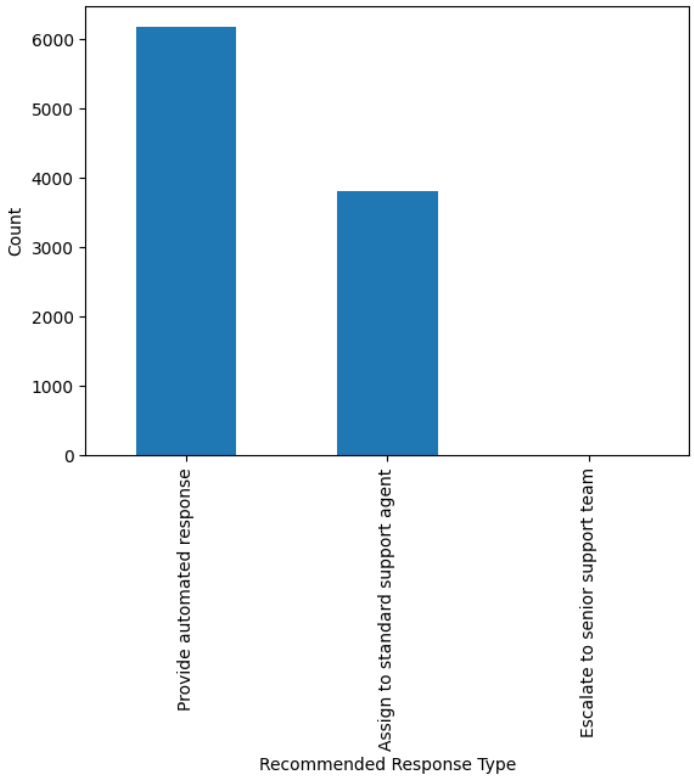


```
          query_type  response_time_min priority_level  \
0  Billing Question               1.88           Low
1    Password Reset               1.09           Low
2  Shipping Inquiry               1.13           Low
3    Account Update               1.12           Low
4  Billing Question               9.85        Medium

               recommended_response
0        Provide automated response
1        Provide automated response
2        Provide automated response
3        Provide automated response
4  Assign to standard support agent
```

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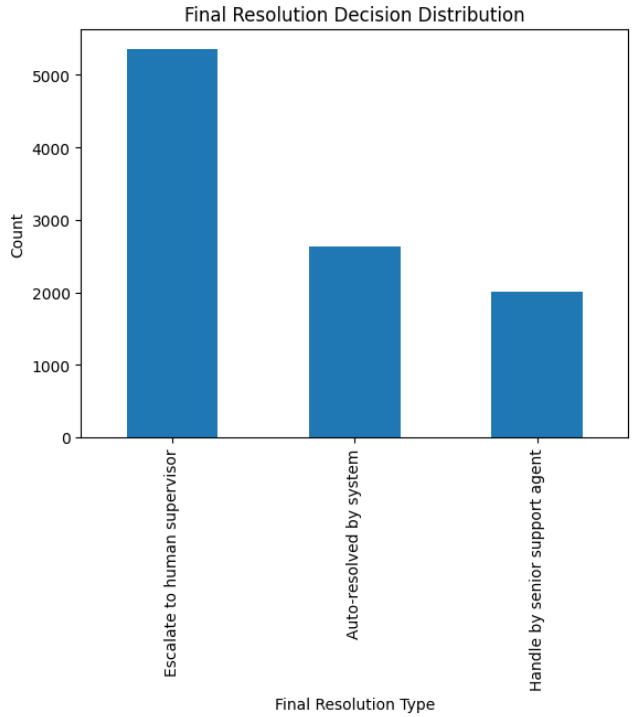


```
         query_type  response_time_min  customer_sentiment  \
0  Billing Question               1.88                0.12
1    Password Reset               1.09                0.14
2  Shipping Inquiry               1.13                0.26
3    Account Update               1.12                0.33
4  Billing Question               9.85                0.87

   follow_up_required priority_level              final_resolution
0               False           Low  Escalate to human supervisor
1               False           Low  Escalate to human supervisor
2               False           Low  Escalate to human supervisor
3                True           Low  Escalate to human supervisor
4                True        Medium  Escalate to human supervisor
```

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